

EVA_OS Review and Handoff Report

Generated: 2026-06-15 Australia/Sydney

This combined report is generated from the handoff review pack files.

EVA_OS 开发进度报告

生成时间：2026-06-15 Australia/Sydney

Executive Summary

EVA_OS 当前是一个本地优先、证据驱动、研究与回测导向的个人复合智能中台。系统已完成从旧命名到 `EVA\_OS` 的统一，量化研究主入口、报告与证据台账、事件驱动行情层、可复现数据湖、事件回放、Token ROI、现金流、政策、消费、报告证

当前状态适合继续开发、审核、演示和个人研究使用，但不应被描述为全自动投资系统或实盘交易系统。系统明确禁止自动实

Current Verified State

Item	Current Status
---	---
Product name	`EVA_OS`
Local source	`/Users/linzezhang/Documents/Codex/2026-06-13/files-mentioned-by-the-user-eva/outputs/CodexFinance`
GitHub repository	`https://github.com/LinzeColin/EVA_OS`
Verified GitHub HEAD	`b9a2c0351efaed332bd6f15b677b32c5835576da`
Local app entries	Desktop, Downloads, Applications: `EVA_OS.app`
Python source files	132
Test files	63
Scripts	40
Markdown docs	32
Latest focused tests	28 passed in 31.48s on 2026-06-13
Full suite	Not claimed as passed

Completed Milestones

Milestone	Status	Evidence
---	---	---
EVA_OS naming and app identity	Complete	`README.md`, `docs/EVA_OS.md`, `macos/EVA_OS.app`, `assets/EVA_OSAppIcon.*`
macOS app entry	Complete	`/Users/linzezhang/Desktop/EVA_OS.app`, `/Users/linzezhang/Downloads/EVA_OS.app`, `/Applications/EVA_OS.app`
Legacy visible app cleanup	Complete	Old visible app bundles were removed in the last verified run
QuantLab daily research loop	Usable	Streamlit app, reports, scripts, docs
Daily readiness / audit artifacts	Implemented	`data/systemAudit`, `docs/DailyReadiness.md`, `docs/DataTrust.md`
Market Event Layer	MVP complete	`src/quantlab/data/market_events.py`, `docs/MarketEventLayer.md`
Reproducible Data Lake	MVP complete	`src/quantlab/data/lake.py`, `docs/ReproducibleDataLake.md`
Event Replay	MVP complete	`src/quantlab/data/replay.py`, `docs/EventReplay.md`

| Token ROI Ledger | Partial product | `src/quantlab/value/token\_roi.py`,
 `docs/TokenROI.md` |
 | Company CashFlow Command | Partial product | `src/quantlab/business/cashflow.py`,
 `docs/CompanyCashFlowCommand.md` |
 | Policy Intelligence Radar | Partial product | `src/quantlab/policy/radar.py`,
 `docs/PolicyIntelligenceRadar.md` |
 | Consumption Guard | Partial product | `src/quantlab/consumption/guard.py`,
 `docs/ConsumptionGuard.md` |
 | Executive Command Center | Partial product |
 `src/quantlab/executive/command\_center.py`, `docs/ExecutiveCommandCenter.md` |
 | 52ETF reference path | Initial integration | `src/quantlab/integrations/site52etf.py`,
 UI warning path |

Current Maturity Assessment

The existing `docs/MaturityRoadmap.md` says maturity is approximately 100%, but current open questions and unfinished work show that overall product maturity should be assessed more conservatively.

Layer	Practical Maturity	Notes
Local app identity and launchers	95%	App entries and icon are in place.
Signing/notarization not productized.		
QuantLab research/backtest workflow	75%	Usable for personal research; full suite
and env setup still need hardening.		
Evidence/reporting layer	70%	Many artifacts exist; real input evidence quality
still varies by subsystem.		
Event/data/replay foundation	65%	MVPs exist; multi-asset replay and cursor
pagination remain.		
Command Center	55%	Aggregates many artifacts; still often outputs `NeedsReview`.
Token ROI / cashflow / policy / consumption	45%	Frameworks exist; real reviewed
evidence refresh is incomplete.		
TradingView-like UX and realtime research	20%	Planned but not stable product
surface yet.		
Real-money execution	0% by design	Explicitly out of scope unless future user
 explicitly changes boundary. |

Last Verified Engineering Result

Latest verified run, based on `HANDOFF.md`:

```
'''text
EVA_OS app plist/icon check: passed for Desktop, Downloads, and Applications
legacy-name scan: no deprecated product-name text or filename residue in scoped project
target report regeneration: Token ROI, Policy Radar, Consumption Guard, Report Decision
Support, and Command Center regenerated
py_compile: passed for renamed app/report/data/integration modules
target pytest: 28 passed in 31.48s
'''
```

Main Risks

Risk	Severity	Current Mitigation
Full test suite not recently completed	Medium	Focused tests passed; full `.venv`
should be rebuilt before broader changes		
Some subsystems use placeholder or missing evidence	Medium	Command Center marks

missing evidence as review-needed |
| Public GitHub data boundary | High | Private holdings, raw imports, SQLite runtime
state, logs, and secrets are excluded |
| Users may overinterpret research outputs as trading advice | High | README and docs
state research-only, no live trading |
| Large Streamlit app file | Medium | Future refactor should split views incrementally |

Recommended Review Questions for ChatGPT

1. Are the safety boundaries sufficient for a personal finance research system?
2. Which subsystem should be prioritized first to increase real decision value?
3. Is the current Token ROI model defensible, or does it need a stricter measurement schema?
4. Are the event replay and data lake contracts stable enough to support vectorized research?
5. Which docs should be shortened or reorganized for lower context cost?
6. What is the minimum acceptance gate before calling CashFlow, Policy, Consumption, and Token ROI fully productized?

EVA_OS 功能清单

System Overview

EVA_OS 是主系统身份，QuantLab 是当前主要可操作入口。系统由 9 个规划子系统和若干共享底座组成。

Main Subsystems

No.	Subsystem	Chinese Name	Current State	Primary Artifacts
01	Executive Command Center	总控驾驶舱	Partial product	'src/quantlab/executive/command_center.py', 'data/commandCenter/*latest*'
02	Token ROI Ledger	Token经济转化台账	Partial product	'src/quantlab/value/token_roi.py', 'data/value/*latest*'
03	Company CashFlow Command	公司经营现金流系统	Partial product	'src/quantlab/business/cashflow.py', 'docs/CompanyCashFlowCommand.md'
04	QuantLab	量化研究与回测系统	Usable core	'src/quantlab/backtest', 'src/quantlab/strategies', Streamlit app
05	Policy Intelligence Radar	政策机会情报系统	Partial product	'src/quantlab/policy/radar.py', 'data/policy/*latest*'
06	Consumption Guard	个人消费止血系统	Partial product	'src/quantlab/consumption/guard.py', 'data/consumption/*latest*'
07	AI Research Engine	AI行业研究系统	Planned / partially represented	Docs and research report paths
08	Sports Market Lab	赛事市场分析系统	Planned / external adjacent	Open questions mention future calibration/reporting
09	CodexForge Factory	Codex工程交付工厂	Planned / workflow layer	Task packs, handoff, packaging conventions

Shared Foundation Layers

Layer	Purpose	Current Implementation
Evidence Layer	Track evidence, source, freshness, and limitation	Docs and generated artifacts use evidence levels

| Data Layer | Normalize and record data quality | Data providers, data quality reports, data lake manifest |
 | Market Event Layer | Convert OHLCV bars into deterministic events |
 `MarketEventLog\_latest.\*` |
 | Reproducible Data Lake | Register immutable local data assets |
 `DataLakeManifest\_latest.\*` |
 | Event Replay | Feed deterministic replay batches to future research engines |
 `EventReplay\_latest.\*` |
Decision Layer	Downgrade conclusions when evidence is missing	Command Center and report decision support
Engineering Layer	Scripts, tests, docs, handoff, repeatable checks	`scripts/\*\*`, `tests/\*\*`, `HANDOFF.md`
Value Layer	Register outputs into Token ROI	`EVATokenROILedger\_latest.\*`

Current User-Facing Functions

Function	What It Does	User Benefit
macOS app launcher	Opens EVA_OS from Desktop, Downloads, or Applications	Faster daily access
Streamlit QuantLab app	Provides research, backtest, holdings, reports, profile, workbench views	Main interactive experience
Single-symbol backtest	Runs strategy tests on selected market data	Strategy exploration
Parameter scan	Compares strategy parameter ranges	Research efficiency
Strategy library	Maintains strategy templates and categories	Repeatable experiments
Report center	Finds and reads generated Word/PDF/Markdown reports	Decision traceability
Data provider checks	Supports sample, Yahoo, AKShare, TuShare, Alpha Vantage, Polygon paths	Data flexibility
Data quality and cross-source validation	Checks freshness and agreement across sources	Lower false-confidence risk
Research bus	Shares local artifacts and messages across systems	Future multi-system orchestration
52ETF reference integration	Initial read-only reference path	External ETF context

Current Generated Latest Artifacts

Artifact Family	Latest Files
Command Center	`data/commandCenter/EVACommandCenter\_latest.json/md/pdf`
Token ROI	`data/value/EVATokenROILedger\_latest.json/csv/md/pdf`
Policy Radar	`data/policy/PolicyIntelligenceRadar\_latest.json/csv/md/pdf`
Consumption Guard	`data/consumption/ConsumptionGuard\_latest.json/csv/md/pdf`
Report Decision Support	
`data/reportDecision/ReportDecisionSupportIndex\_latest.json/csv/md/pdf`	
Report Gap Tasks	`data/reportDecision/ReportEvidenceGapTasks\_latest.json/csv/md/pdf`
Market Events	`data/marketEvents/MarketEventLog\_latest.json/jsonl/csv/md`
Data Lake	
`data/dataLake/DataLakeManifest\_latest.json/md/assets.csv/replay\_cursors.json`	
Event Replay	`data/replay/EventReplay\_latest.json/csv/md`
Validation Queue	`data/validationQueue/\*latest\*`

Explicit Non-Functions

EVA_OS currently does not do these things:

- No autonomous real-money trading.
- No real broker order placement.
- No automatic betting.
- No automatic payments, tax filings, or government submissions.
- No storing brokerage passwords or secrets in source.
- No guarantee that generated research is investment advice.
- No claim that all external data provider paths are currently authenticated or live-validated.

新开发者交接说明

First Rule

Do not start by changing code. First read the current source of truth:

```
'''text
README.md
HANDOFF.md
AGENT_CONTINUITY.md
15_OPEN_QUESTIONS.md
UPLOAD_MANIFEST.md
docs/EVA_OS.md
docs/Index.md
'''
```

Local Project

```
'''text
/Users/linzezhang/Documents/Codex/2026-06-13/files-mentioned-by-the-user-eva/outputs/CodexFinance
'''
```

GitHub:

```
'''text
https://github.com/LinzeColin/EVA_OS
HEAD: b9a2c0351efaed332bd6f15b677b32c5835576da
'''
```

Environment Notes

The old ``.venv`` was removed during slimming. Some scripts now fall back to ``QUANTLAB_PYTHON`` or system ``python3``, but broad development should recreate a full environment.

Recommended setup:

```
'''bash
cd /Users/linzezhang/Documents/Codex/2026-06-13/files-mentioned-by-the-user-eva/outputs/CodexFinance
python3 -m venv .venv
.venv/bin/python -m pip install -e '[test,app,data]'
```

'''

Safe Verification Commands

Focused checks used recently:

```
'''bash
PYTHONPATH=src .venv/bin/python -m pytest \
  tests/test_command_center.py \
  tests/test_cashflow_command.py \
  tests/test_policy_radar.py \
  tests/test_consumption_guard.py \
  tests/test_market_event_layer.py \
  tests/test_data_lake_manifest.py \
  tests/test_event_replay.py \
  -q -p no:cacheprovider
'''
```

If `.venv` is unavailable, do not claim the tests passed. Either rebuild `.venv` or document exactly which runtime was used.

Architecture Map

Area	Paths
UI	<code>src/quantlab/app/streamlit_app.py</code> , <code>src/quantlab/app/dashboard.py</code>
Backtest and strategies	<code>src/quantlab/backtest</code> , <code>src/quantlab/strategies</code> , <code>src/quantlab/indicators</code>
Data providers and quality	<code>src/quantlab/data</code> , <code>src/quantlab/data/providers</code>
Market events / data lake / replay	<code>src/quantlab/data/market_events.py</code> , <code>src/quantlab/data/lake.py</code> , <code>src/quantlab/data/replay.py</code>
Reports	<code>src/quantlab/reports</code> , <code>docs/ReportGuide.md</code>
Executive command center	<code>src/quantlab/executive/command_center.py</code>
Token ROI	<code>src/quantlab/value/token_roi.py</code>
Business systems	<code>src/quantlab/business</code> , <code>src/quantlab/policy</code> , <code>src/quantlab/consumption</code>
External integrations	<code>src/quantlab/integrations/site52etf.py</code>
System identity and health	<code>src/quantlab/system/eva_identity.py</code> , <code>src/quantlab/system/health.py</code>
Scripts	<code>scripts/**</code>
Tests	<code>tests/**</code>

Development Rules

1. Work on one subsystem at a time.
2. Preserve research-only safety boundaries.
3. Do not upload private holdings, screenshots, SQLite runtime state, secrets, `.env`, logs, or raw imports.
4. Update docs and tests when behavior changes.
5. Push meaningful continuity changes to GitHub.
6. Keep `HANDOFF.md` short and factual.
7. Avoid broad refactors of the 7,000+ line Streamlit app unless the target run is specifically UI modularization.

Current Highest-ROI Next Run

Default next run: `'Vectorized Research Mode MVP'`.

Scope:

1. Read `data/replay/EventReplay\_latest.json`.
2. Convert replay events to a stable OHLCV DataFrame.
3. Expose a fast parameter-scan adapter.
4. Add focused tests.
5. Keep it read-only and deterministic.
6. Do not add realtime trading or broker execution.

Definition of Done for New Work

A future change should not be considered done until:

- code is implemented in the smallest relevant modules;
- docs are updated;
- focused tests pass;
- safety boundaries are unchanged or stronger;
- generated artifacts are refreshed only when relevant;
- `HANDOFF.md` is updated if continuity changed;
- GitHub push is complete when the user asked for sync.

EVA_OS 任务清单与开发路线图

Work Sequencing Principle

Do not upgrade all systems at once. Continue one subsystem per run to reduce confusion, testing risk, and token/context pressure.

Current Open Work

Priority	Task	Goal	Suggested Acceptance
---	---	---	---
P0	Rebuild full dev environment	Restore reliable full-suite validation	`venv` exists, dependencies installed, smoke tests pass
P0	Vectorized Research Mode MVP	Use replay batches for fast deterministic research	Replay JSON -> OHLCV DataFrame -> parameter scan test
P0	Hotspot analysis performance	Reduce slow button-trigger behavior	Cached/precomputed path, user-visible latency reduced
P1	Replay cursor pagination	Correct replay when timestamps collide	Cursor includes timestamp + event_id
P1	Bar cache replay	Replay from structured cache files	CSV/parquet input covered by tests
P1	Multi-asset replay	Synchronize multiple symbols/intervals	Deterministic merge order and tests
P1	Workbench consolidation	Reduce context/token pressure	Summary-first command surface and fewer overlapping views
P1	52ETF integration	Make read-only ETF reference path useful	Source log, cache, failure mode, docs
P2	Discrete Event Simulation Mode	Event-by-event simulation with costs/risk	Portfolio state, slippage/cost model, audit log
P2	Agent Market Simulation Mode	Research-only agent loop	No real orders, deterministic input, audit output
P2	TradingView-like UX	Chart/indicator/strategy controls	Stable core first, then

UI |
P2	Moomoo realtime research	Read-only OpenD quote ingestion	Fail-closed readiness, no trading
P2	Business evidence completion	Productize cashflow/policy/consumption	Real reviewed evidence snapshots and action gates
P3	Formal audit pack regeneration	Create new `eva\_os\_dev\_audit\_pack`	Only after next stable subsystem run

Suggested Iteration Plan

Iteration 1: Environment and Smoke Gate

Goal: make future development repeatable.

Steps:

1. Recreate `.venv`.
2. Install test/app/data extras.
3. Run focused tests.
4. Split test commands into smoke, target, integration, and full.
5. Update `docs/Testing.md`.

Iteration 2: Vectorized Research Mode MVP

Goal: turn Event Replay into practical research input.

Steps:

1. Add replay-to-OHLCV conversion module.
2. Add DataFrame validation rules.
3. Add parameter scan adapter.
4. Add tests for sample replay.
5. Add CLI smoke script.
6. Update docs.

Iteration 3: Hotspot Analysis Optimization

Goal: reduce slow UI actions and token pressure.

Steps:

1. Profile current hotspot button path.
2. Identify repeated expensive calls and generated text paths.
3. Add cache/precompute layer.
4. Add progress/status UI.
5. Validate no stale or fabricated conclusion.

Iteration 4: 52ETF Productization

Goal: convert 52ETF from initial integration into a reviewed research reference.

Steps:

1. Define allowed data and request boundaries.
2. Add source log and cache.
3. Add freshness and failure-state display.

4. Add tests with mocked response.
5. Add user guidance and risk note.

Iteration 5: Business Subsystems Evidence Gate

Goal: move Token ROI, CashFlow, Policy, and Consumption closer to fully productized state.

Steps:

1. Define evidence input templates.
2. Add manual review queue.
3. Add missing evidence log.
4. Add value and decision records.
5. Update Command Center scoring.

Stop Conditions

Stop and report instead of expanding scope when:

- full environment cannot be built;
- tests fail due to unclear dependency mismatch;
- a feature would require private credentials or external account access;
- a requested change risks enabling live trading or real payments;
- evidence is missing but output would look actionable.

EVA_OS 使用者说明简版

What EVA_OS Is

EVA_OS 是个人研究与决策支持中台。它帮助整理市场数据、策略回测、报告、证据、现金流、政策、消费和 Token ROI，不

What You Can Use Today

Use Case	How
打开系统	双击 `EVA\_OS.app`
查看总控状态	进入总控驾驶舱或运行 `scripts/commandCenter.sh`
运行策略回测	使用 QuantLab 的单标的回测 / 策略入口
做参数扫描	使用参数扫描功能或相关脚本
查看报告	进入报告中心或查看 `data/\*\*/\*latest\*`
查看 Token ROI	查看 `data/value/EVATokenROILedger\_latest.\*`
查看政策/消费/现金流快照	查看对应 docs 和 `data/policy`, `data/consumption`, `data/cashflow`
查看系统文档	先读 `README.md`, `docs/QuickStart.md`, `docs/Index.md`

App Entry Points

```
'''text
/Users/linzezhang/Desktop/EVA_OS.app
/Users/linzezhang/Downloads/EVA_OS.app
/Applications/EVA_OS.app
'''
```

Safety Boundary

EVA_OS will not:

- place real trades;
- connect to broker execution by default;
- automatically bet;
- automatically pay;
- store brokerage passwords;
- replace human judgment.

Treat outputs as research evidence and review material, not as final financial advice.

Recommended Daily Flow

1. Open `EVA\_OS.app`.
2. Check Command Center status.
3. Review missing evidence and action queue.
4. Run backtests or reports only on the needed target.
5. Save important outputs into the report/evidence flow.
6. Do not act on results until evidence quality and limitations are reviewed.

What Still Needs Work

The system still needs more productization in:

- faster hotspot analysis;
- cleaner consolidated workbench;
- better 52ETF reference ingestion;
- vectorized research mode;
- realtime read-only research flow;
- stronger real evidence inputs for Token ROI, CashFlow, Policy, and Consumption.

EVA_OS 审核评估清单

This file is intended for ChatGPT, reviewers, and developers auditing the system.

Review Objectives

1. Check whether current scope and safety boundaries are clear.
2. Identify implementation gaps before product claims.
3. Prioritize high-ROI next development tasks.
4. Reduce token/context cost for future agents.
5. Confirm public GitHub boundary is safe.

Product Review Checklist

Area	Question	Current Evidence	Review Result
Naming	Is every visible product name `EVA_OS`?	Last scan reported zero deprecated-name residue	Pending reviewer confirmation
User entry	Can users launch the app easily?	Desktop, Downloads, Applications app entries	Pending live user test

Core value	Does the system support actual research decisions?	Backtest, reports, evidence, command center	Partial
Safety	Does the system prevent live execution risk?	README and docs prohibit real trading/payment	Good, keep monitoring
Evidence quality	Are conclusions evidence-backed?	Evidence levels exist, but real inputs are incomplete	Needs improvement
Test health	Are checks repeatable?	28 focused tests passed; full suite not claimed	Needs full env
Data privacy	Is public GitHub safe?	Upload manifest excludes private paths	Review before every push

Code Review Checklist

Area	What to Check
Schema stability	`EVAOS\*` schema names and generated data compatibility
Streamlit app size	`src/quantlab/app/streamlit\_app.py` is very large; avoid broad edits
Script runtime	Scripts should consistently support `.venv`, `QUANTLAB\_PYTHON`, or documented fallback
Data paths	Avoid hard-coded historical paths
External requests	52ETF and provider calls should be cached, source-logged, and fail closed
Generated artifacts	Avoid committing private, large, or stale generated outputs
Tests	Add target tests for any changed subsystem

Security and Privacy Checklist

Risk	Required Rule
Secrets	Never commit `.env`, API keys, tokens, passwords
Holdings	Do not upload private holdings or account-derived raw files
Runtime DB	Do not upload SQLite runtime state unless explicitly sanitized
Logs	Do not upload local logs, pids, locks
Broker execution	No real order submission without explicit future authorization and separate approval design
User content	Treat local screenshots/videos/imports as private

Acceptance Criteria for “ Productized ”

A subsystem can be called productized only when:

- it has clear user purpose and workflow;
- it has documented input schema;
- it has source/evidence logging;
- it has missing-data handling;
- it has generated outputs in JSON/CSV/MD/PDF where appropriate;
- it has focused tests;
- it has UI or CLI access;
- it has risk boundaries;
- it appears in Command Center status;
- it has a meaningful user-facing interpretation, not just raw data.

Suggested Reviewer Output Format

Ask ChatGPT or another reviewer to return:

1. Top 10 risks.
2. Top 10 high-ROI improvements.
3. Which subsystem to do next and why.
4. What should not be built yet.
5. Any misleading wording in docs.
6. Minimum test gate before next GitHub push.